

Inflation is Conflict

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Introduction

- This paper argues that the **most proximate and general cause of inflation** is **conflict** (disagreement over relative prices).
- In this view, inflation results from **incompatible goals (aspirations)** over relative prices by economic agents.
 - Due to **nominal rigidities**, agents occasionally change their prices.
 - When they do, they adjust them to **influence relative prices** according to their **aspiration**.
 - **Staggered prices** coupled with **conflict**, that is, disagreement over relative prices, leads to a finite level of inflation.

Introduction

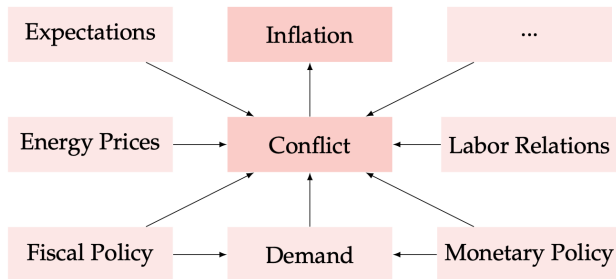


Figure: Inflation DAG: Conflict as Proximate with other Root Causes

- The goal of this paper is to provide a framework to think about disagreement over relative prices as the proximate cause of inflation.
- Specific models may then be thought of as endogenizing conflict in different ways. We will **not** do that here.

Outline

- **Simple Model:** Inflation from Conflict without Money.
- **General Framework:** Inflation and Conflict in a Network Economy
 - Defining Conflict
 - A Framework for Price Dynamics
 - Showing that Inflation equals Conflict
- **Conclusion**

Simple Model

Technology and Preferences

- **Two agents:** A and B, and two non-storable goods, also labeled A and B.
- Time is **discrete**. Each period, agents have an endowment (normalized to 1) of their respective good: A owns A, B owns B.

- Utility:

$$U^A = \sum_{t=0}^{\infty} \beta^t u(c_t^A, c_t^B)$$

$$U^B = \sum_{t=0}^{\infty} \beta^t u(c_t^B, c_t^A),$$

$\beta < 1$. Preferences within a period are **symmetric**: $u(c, c')$, where c denotes consumption of their own good and c' denotes consumption of the other good.

- u is concave, twice differentiable and strictly increasing in both arguments.
- There is no way to save or borrow.

Staggered Prices

- Prices expressed in a common unit of account.
- Prices are set in a **staggered** fashion and remain unchanged for two periods: Agent **A** sets prices in **even** periods and agent **B** in **odd** periods.
- P_t^* denotes a **newly reset price**:

$$P_t^* = P_t^A = P_{t+1}^A \quad \text{for } t = 0, 2, \dots$$

$$P_t^* = P_t^B = P_{t+1}^B \quad \text{for } t = 1, 3, \dots$$

- The price of B at $t = 0$ is given.
- Using the above conditions, the sequence of reset prices $\{P_t^*\}$ determines all prices in this economy $\{P_t^A, P_t^B\}$
 $\implies$ Our goal is to solve for the **equilibrium sequence** $\{P_t^*\}$.

Trade by Barter

- In each period t there will be a “seller” (agent who resets their price) and a “buyer” (agent who does not get to reset their price).
- All trade takes place by **barter** using as **terms of trade** the ratio of the currently posted prices.

	Seller	Buyer	Relative Price of A
t even	A	B	$\frac{p_t^A}{p_t^B} = \frac{p_t^*}{p_{t-1}^*}$
t odd	B	A	$\frac{p_t^A}{p_t^B} = \frac{p_{t-1}^*}{p_t^*}$

- Quantities** are determined by a take-it-or-leave-it offer proposed by the buyer that the seller can **accept** or **reject**.

Trade by Barter in more detail

- In **even** periods, after A has reset their price, **agent B acts as the buyer** and offers to buy c_t^A units of good A and pay

$$\frac{P_t^A}{P_t^B} c_t^A = \frac{P_t^*}{P_{t-1}^*} c_t^A$$

units of good B.

- Agent A can decide whether to accept or reject this offer.
- If the offer is
 - **accepted**, they execute the barter exchange and consume.
 - **rejected**, both agents consume their endowment in that period.
- In odd periods the trading protocol is identical, but reversing the roles of A and B.

Equilibrium

- We want to show that the equilibrium in this economy will have inflation.
- We focus on **Markov equilibria**, that is, equilibria that depend on the relevant **state variables**:
 - P_{t-1}^* before the current price is set.
 - (P_{t-1}^*, P_t^*) after the current price is set.
- We solve backwards, **starting with the buyer problem**, then turning to the seller problem setting its price.

Buyer Problem

- In a given period t , **after prices are set**, the buyer can buy the good of the seller at the relative price

$$p_t = \frac{P_t^*}{P_{t-1}^*}.$$

If t is even, p_t is the price of good A in terms of B.

If t is odd, it is the price of good B in terms of A.

- Dropping time subscripts, the **buyer solves**

$$V(p) = \max_{c, c'} u(c, c')$$

$$\text{s.t. } c = 1 - pc'$$

$$u(1 - c', pc') \geq u(1, 0).$$

- The **solution** can be written as a demand schedule:

$$c' = D(p), \quad c = 1 - pD(p)$$

Seller Problem

- By varying P_t^* the seller is able to freely determine the relative price $p_t = P_t^* / P_{t-1}^*$. Therefore, seller's problem is **static** and given by:

$$p^* \equiv \arg \max_p u((1 - D(p), pD(p))).$$

- The first order condition gives

$$p^* = \underbrace{\frac{1}{1 - 1/\varepsilon(p^*)}}_{\text{mark-up}} \cdot \underbrace{\frac{u_c(c, c')}{u_{c'}(c, c')}}_{MRS},$$

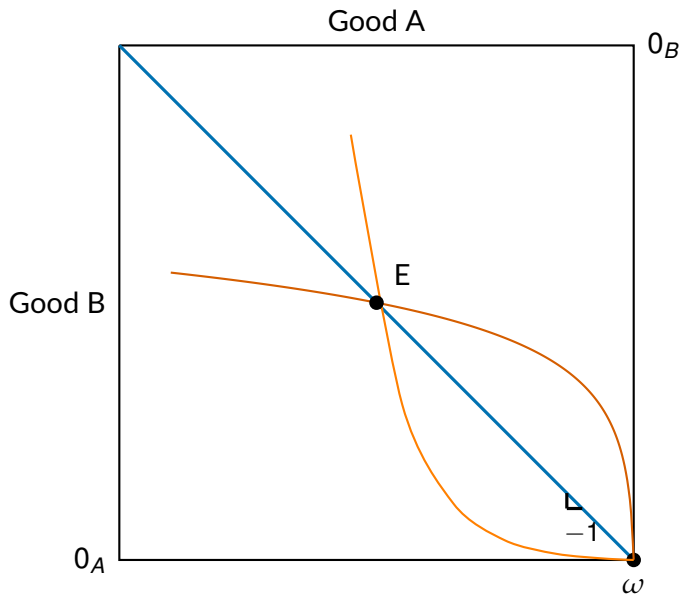
where $c = 1 - D(p)$, $c' = pD(p)$ and $\varepsilon(p)$ denotes the local demand elasticity $-D'(p)p/D(p)$.

- We want to see that $p^* > 1$. That is, that there is inflation.

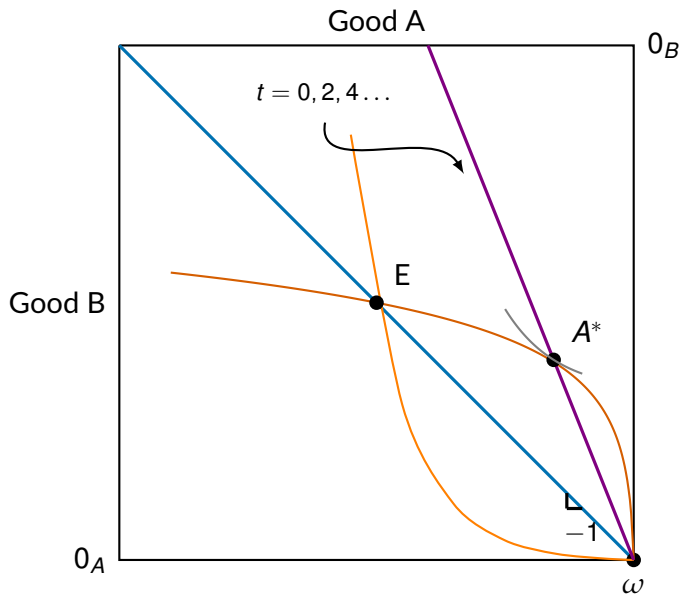
Competitive Equilibrium Benchmark

- This is an exchange economy, repeating without change each period. Allocations can be pictured in an Edgeworth-box diagram.
- This model features staggered pricing and market power, but the **competitive equilibrium (CE)** provides a useful benchmark.
- To show $p^* > 1$, we look at the **CE benchmark**.
- By symmetry of preferences and endowments, a symmetric equilibrium must exist, with a **relative price of unity**.
- How does it compare to the equilibrium in the staggered price economy?

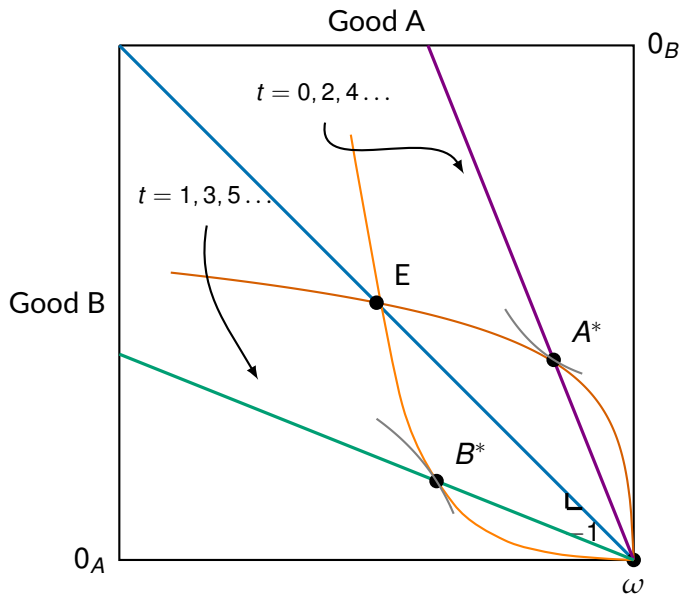
Equilibrium Inflation



Equilibrium Inflation



Equilibrium Inflation



Equilibrium Inflation

- We have seen that the rate of change in nominal prices is **constant** and entirely determined by **preferences and endowments**.
- The seller always optimally chooses a relative price that is **strictly above** that of the competitive equilibrium.
- Since there is a symmetric competitive equilibrium with unity relative price $\implies P_t^* / P_{t-1}^* > 1$.
- This positive inflation comes from **disagreement over relative prices**.

Proposition 1: Inflation is constant and positive, unaffected by β and given by

$$\frac{P_t^*}{P_{t-1}^*} - 1 = \pi^* > 0$$

► Proof

Quasilinear Iso-elastic Example

Simple example in closed form.

- Utility function:

$$u(c, c') = c + d^{\frac{1}{\varepsilon}} \frac{(c')^{1-\frac{1}{\varepsilon}}}{1-\frac{1}{\varepsilon}} \quad \text{with } \varepsilon > 1 \text{ and } d \in (0, 1).$$

- This yields demand curve $D(p) = dp^{-\varepsilon}$, with constant elasticity over $p \geq 1$.
- We get

$$p^* = \frac{1}{1-1/\varepsilon} d^{-\frac{1}{\varepsilon}} (p^* D(p^*))^{\frac{1}{\varepsilon}}$$

which can be solved for

$$p^* = \left(\frac{1}{1-1/\varepsilon} \right)^{\frac{1}{2-1/\varepsilon}} > 1$$

- Inflation is strictly positive.

The Role of Conflict and Staggering

- In this model we get **inflation** -a persistent and generalized rise in nominal prices- in an economy without:
 - Money
 - Credit or savings
 - Interest rates
 - Production
 - Employment
- Inflation results from **conflict** over relative prices and staggered setting of nominal prices.

General Framework:

Inflation and Conflict in a Network
Economy

General Framework - Not a Full Model!

- **Framework:** Subset of the equations of a specific and full model.
- General framework with **staggered price setting** and any number of **interdependent sectors**.
- Standard NKM, for example, fits into this framework.
- We will have **three big results**:
 - 1) No conflict $\iff$ “average” aspirations are zero.
 - 2) Generalized sectoral inflation is conflict.
 - 3) Long-run inflation for every sector is conflict.
- We will study an N sector network economy, $n = 1, 2, \dots, N$.
- We will take as given the aspirations of each sector for relative prices.

The General Network Economy

- P_n is the log price in sector n .
- Each sector cares about relative prices

$$P_n = \sum_{n'} m_{nn'} P_{n'},$$

where $\sum_{n'} m_{nn'} P_{n'}$ is a relevant **price index** for sector n with price index coefficients $m_{nn'} \geq 0$ and $\sum_{n'} m_{nn'} = 1$.

- Let $M = (m_{nn'})$ be the matrix that stacks the **price index coefficients** of each sector in its rows. It has non-negative entries and rows that add up to one.
- M encapsulates the **interdependencies** across sectors in the economy.

Defining Conflict

- For each sector n , let $a_n \in \mathbb{R}$ be sector n 's **aspiration** for their relative price. That is, sector n would like

$$P_n - \sum_{n'} m_{nn'} P_{n'} = a_n$$

- For a given $a = (a_1, a_2, \dots, a_N)$, are these aspirations always **collectively feasible**?

Definition: We say there is **conflict** if there exists no price vector $P \in \mathbb{R}^N$ such that

$$P_n - \sum_{n'} m_{nn'} P_{n'} = a_n \quad \text{for all } n.$$

Conversely, if this condition holds for some P , then we say there is no conflict.

Defining Conflict

Definition (Network Centrality): We define network centrality as γ , the left eigenvector of M with eigenvalue 1. It is a measure of the overall importance of sectors. We will use it as a weight to define relevant averages or indices.

The uniqueness of γ is ensured by assuming that M is irreducible.

Result 1: No Conflict Lemma. *There is no conflict if and only if*

$$\sum_n \gamma_n a_n = 0,$$

*that is, when average aspirations, weighted by **network centrality**, are zero.*

- We adopt $\sum_n \gamma_n a_n$ as a quantitative measure of conflict.
- Our measure for conflict will depend on the **interdependencies** between sectors, and this is captured by weighting with γ .

Wage-Price Example

- $N = 2, n \in \{P, W\}$
- Firms produce consumption goods with nominal price P (in logs).
- Households provide labor to firms at nominal wage W (in logs).
- Let $a = (a_P, a_W)'$ be the aspirations for firms and workers, respectively.
- Aspiration for the firm is given by their desired profit margin:

$$a_P = P - W$$

- Aspiration for the workers is given by their desired real wage:

$$a_W = W - P$$

- Network centrality is $\gamma = [0.5, 0.5]'$.

Conflict in the Wage-Price Example

- There is no conflict if and only if $a_P + a_W = 0$.
- When $a_P + a_W \neq 0$, aspirations are in conflict: $(W - P)$ desired by the workers doesn't coincide with $(W - P)$ desired by the firms.
- $a_P + a_W$ is a measure of conflict: Disagreement over $(W - P)$.
- Here we take a_P and a_W as exogenously given. In a NK model, for example, we could have:

$$a_W = MRS + \text{Union Markup} + \pi^e$$

$$a_P = \text{Firm Markup} - MPL + \pi^e$$

- **Next step:** develop a standard price setting framework that allows us to show that conflict plays a key role in inflation.

Price Dynamics in the general Framework

Environment

- Time is continuous: $t \geq 0$.
- Within each sector $n = 1, 2, \dots, N$ there are a continuum of symmetric varieties $i \in [0, 1]$.
- Nominal prices P_{nit} are in logs.
- The price of sector n in period t is the average price over varieties:

$$P_{nt} = \int P_{nit} di$$

- $P_t = (P_{1t}, \dots, P_{Nt})'$.
- Price index coefficients $(m_{nn'})$ and aspirations $\{a_t\}_{t \geq 0}$ are exogenous.

Pricing Frictions

- Prices of varieties are reset infrequently à la Calvo with Poisson rate λ_n , with independent realizations across time and varieties.
- Evolution of the average price in sector n :

$$\dot{P}_{nt} = \lambda_n (P_{nt}^* - P_{nt}),$$

where P_{nt}^* is the price of varieties in sector n whose prices are reset at t .

- We postulate that the **reset price** is:

$$P_{nt}^* = a_{nt} + \sum_{n'} m_{nn'} P_{n't}$$

for a given path $\{a_{nt}\}_{t \geq 0}$ of aspirations.

$$\implies \pi_{nt} = \dot{P}_{nt} = \lambda_n \left(a_{nt} + \sum_{n'} m_{nn'} P_{n't} - P_{nt} \right)$$

Conflict Inflation

- The last equation already tells us that a particular sector n will have inflation only if they are not attaining their aspirations:

$$\pi_{nt} = \dot{P}_{nt} = \lambda_n \left[a_{nt} - \left(P_{nt} - \sum_{n'} m_{nn'} P_{n't} \right) \right]$$

- We want to see what happens to inflation in an **aggregate price index**.

- Define the aggregate price index $\bar{P}_t = \sum_n \psi_n P_{nt}$, with $\psi_n = \frac{d_n}{\bar{d}} \gamma_n$,

where $d_n = 1 / \lambda_n$ is the price duration of sector n and $\bar{d} = \sum_n \gamma_n d_n$ is average price duration, weighted by centrality.

Conflict Inflation

Definition: (Conflict Inflation). Given network centrality γ and average duration $\bar{d}$, define **conflict inflation** by

$$\Pi_t^C = \frac{1}{\bar{d}} \sum_n \gamma_n a_{nt}.$$

Result 2: Generalized sectoral inflation is conflict

Inflation in the aggregate price index equals conflict inflation:

$$\bar{\pi}_t = \sum_n \psi_n \pi_{nt} = \Pi_t^C$$

Moreover, there exists a unique relative price vector $\mathcal{P}(a)$ such that if $p_t \equiv P_t - \bar{P}_t = \mathcal{P}(a_t)$, all sectors have inflation equal to conflict inflation:

$$\pi_{nt} = \Pi_t^C, \quad n = 1, 2, \dots, N.$$

General Framework - Implications

- Intuition:

$$\dot{p}_t = \underbrace{\dot{P}_t}_{\pi_t} - \underbrace{\dot{\bar{P}}_t}_{\Pi_t^C} \implies \pi_t = \Pi_t^C + \text{"Adjustment in relative prices"}.$$

- Zero average inflation holds if and only if conflict is absent:

$$\bar{\pi}_t = 0 \iff \sum_n \gamma_n a_{nt} = 0.$$

- Conflict inflation depends only on **current aspirations** a_t .
Therefore, positive inflation requires current conflict in aspirations.

Inflation over the Long-Run

Result 3: Long-Run Inflation is Conflict. Suppose the path for aspirations $\{a_t\}$ is bounded. Then long-run average inflation equals long-run average conflict inflation:

$$\bar{\pi}_n = \bar{\Pi}^C$$

for all sectors n , where

$$\bar{\pi}_n = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \pi_{nt} dt \quad \text{and} \quad \bar{\Pi}^C = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \Pi_t^C dt$$

► Proof

- Therefore any prolonged episodes of inflation must be driven by conflict.
- Inflation is often defined as a **sustained and generalized** increase in prices.
 $\implies$ **Inflation is conflict.**

Conflict Inflation in the Wage-Price Example

- Price setting and wage setting are

$$\begin{aligned}P_t^* &= a_{Pt} + W_t \\ W_t^* &= a_{Wt} + P_t,\end{aligned}$$

so that

$$\begin{aligned}\pi_t^P &= \lambda_P(P_t^* - P_t) = \lambda_P(a_{Pt} + (W_t - P_t)) \\ \pi_t^W &= \lambda_W(W_t^* - W_t) = \lambda_W(a_{Wt} - (W_t - P_t))\end{aligned}$$

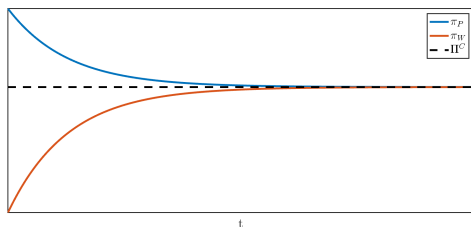
- Conflict inflation is

$$\Pi_t^C \equiv \frac{1}{\bar{d}} \sum_n \gamma_n a_n = \frac{\lambda_W \lambda_P}{\lambda_W + \lambda_P} (a_{Pt} + a_{Wt}).$$

Conflict Inflation in the Wage-Price Example

$$\Pi_t^C = \frac{\lambda_W \lambda_P}{\lambda_W + \lambda_P} (a_{Pt} + a_{Wt}).$$

- If $\lambda_W = 0$ or $\lambda_P = 0 \implies \Pi_t^C = 0$. Conflict inflation is amplified by the **feedback across sectors**.
- If $\lambda_W \rightarrow \infty \implies \Pi_t^C = \lambda_P (a_{Pt} + a_{Wt})$
- Example: $a_{Wt} = a_{Pt} = 0$ for all $t < 0$. For $t \geq 0$, $a_P > 0$:



Conclusion

Conclusion

- This paper provides a perspective on the **role of conflict** -defined as disagreement on relative prices- **on inflation**.
 - Individuals make efforts to adjust the absolute prices under their control to meet these relative price goals (aspirations).
 - If their goals are **collectively incompatible** these efforts are destined to fail, and set in motion a trend in absolute prices.
 - Conflict over relative prices translates into inflation.
- The view in this paper is that the **more proximate cause** of inflation is always conflict. However, conflict is typically endogenous.
- This is a **General framework** for thinking about inflation that nests many traditional models, but can potentially be much broader.

Thank you!

Appendix

Non-Symmetric Preferences

- Now agent A has utility $u_A(c_A, c_B)$ and agent B has utility $u_B(c_A, c_B)$, **we no longer impose** $u_A(c, c') = u_B(c', c)$.
- Take any competitive equilibrium relative price for the static economy:
 $\hat{p} = \frac{P_A}{P_B}$
- Using the same logic as in Prop. 1, when A resets their price, they will ensure a relative price that is higher than the CE, and symmetrically when B resets their price:

$$\frac{P_t^*}{P_{t-1}^*} = p_A^* > \hat{p} \quad \text{and} \quad \frac{P_t^*}{P_{t-1}^*} = p_B^* > \frac{1}{\hat{p}}$$

- Combining the two inequalities:

$$\frac{P_{t+1}^*}{P_t^*} \frac{P_t^*}{P_{t-1}^*} = \frac{P_{t+1}^*}{P_{t-1}^*} = p_A^* p_B^* > 1 \quad \text{for } t = 0, 1, \dots$$

Forward-looking Aspirations

- We saw that sectors that reset their prices obtained their relative price equal to their aspirations a_{nt} .
- a_t captures the aspirations for the relative price obtained **at that moment**. These aspirations are immediately realized.
- With positive inflation these relative prices will not be sustained and will deteriorate over time.
- One possible interpretation of the aspirations $\{a_t\}$ is that they represent **naïve goals** that are frustrated by positive inflation.
- This is not the preferred interpretation in this paper. The results we've shown hold even when $\{a_t\}$ are endogenous.
- We now explore aspirations that incorporate **inflation expectations**. This amounts to a first step in endogenizing expectations.

Aspirations Adjusted by Expected Inflation

- We consider forward looking agents that have some desired targets $\{a_{nt}\}$ for their relative price at time t .
- We postulate that now the reset price satisfies

$$P_{nt}^* = (\rho + \lambda_n) \hat{\mathbb{E}}_{nt} \int_t^\infty e^{-(\rho + \lambda_n)(s-t)} \left(\hat{a}_{ns} + \sum_{n'} m_{n'n} P_{n's} \right) ds,$$

- This reset equation is consistent with $P_{nt}^* = a_{nt} + \sum_{n'} m_{nn'} P_{n't}$ with an endogenous reset aspiration given by:

$$a_{nt} = \hat{\mathbb{E}}_{nt} \int_t^\infty e^{-(\rho + \lambda_n)(s-t)} \left((\rho + \lambda_n) \hat{a}_{ns} + \sum_{n'} m_{nn'} \pi_{n's} \right) ds$$

- This shows **expected inflation raises aspirations** $\{a_{nt}\}$ for a given path of target aspirations, creating an additional feedback channel across sectors.
- Sectors are concerned with inflation in other sectors according to their own price index. [▶ Back](#)

Simple Example to Fix Ideas

- Consider a simple case with constant target aspirations $\hat{a}_{nt}$ and constant inflation expectations $\hat{\mathbb{E}}_{nt}[\pi_{n't+s}] = \pi_{nn'}^e$. Then,

$$a_n = \hat{a}_n + \frac{\sum_{n'} m_{nn'} \pi_{nn'}^e}{\rho + \lambda_n}$$

- Price must be reset more aggressively to attain the same targets if expected inflation is positive.
- Computing our measure of conflict:

$$\sum_n \gamma_n a_n = \underbrace{\sum_n \gamma_n \hat{a}_n}_{\text{target conflict}} + \underbrace{\sum_n \left(\frac{\sum_{n'} m_{nn'} \pi_{nn'}^e}{(\rho + \lambda_n)} \right)}_{\text{inflation expectations of all sectors}} .$$

- Thus, higher expected inflation raises inflation **through conflict**.

Proof of Proposition 1

- Seller cannot choose quantities. If they could, they would obtain

$$\tilde{v}(p) \equiv \max_{c, c'} u(c, c') \quad \text{s.t.} \quad c' = p(1 - c)$$

- Since $(c, c') = (1 - D(p), pD(p))$ is feasible, we have

$$v(p) = u(1 - D(p), pD(p)) \leq \tilde{v}(p) \quad \text{for all } p > 0$$

with strict inequality as long as $(c, c') = (1 - D(p), pD(p))$ is not optimal for $\tilde{v}(p)$.

- $p = 1$ is a CE price and at a CE price, when both agents are price takers they choose quantities consistent with market clearing.

$$\implies v(1) = \tilde{v}(1).$$

- $\tilde{v}(p)$ is strictly increasing in p since $c' > 0$ and thus $1 - c > 0$. We conclude that for any $p < 1$,

$$v(p) \leq \tilde{v}(p) < \tilde{v}(1) = v(1),$$

so $p < 1$ is not optimal for the seller. By the Envelope theorem,

$$v'(1) = \tilde{v}'(1) > 0. \implies p = 1 \text{ cannot be optimal} \implies p^* > 1. \quad \text{Back}$$

Proof of Result 2

- First we prove that inflation in the aggregate price index equals conflict inflation:

$$\begin{aligned}\pi_t &= \dot{P}_t = \Lambda(a_t - AP_t) \\ \underbrace{\psi' \pi_t}_{\bar{\pi}_t} &= \psi' \Lambda a_t - \psi' \Lambda A P_t,\end{aligned}$$

and since $\psi' \Lambda = \frac{1}{\bar{d}} \gamma'$,

$$\bar{\pi}_t = \underbrace{\frac{1}{\bar{d}} \gamma' a_t}_{\Pi_t^C} - \frac{1}{\bar{d}} \underbrace{\gamma' A}_0 P_t$$

$$\implies \bar{\pi}_t = \Pi_t^C$$

Proof of Result 2 - Continued

- Now for the second part of the proof, we want to show that there exists a unique relative price vector $\mathcal{P}(a)$ such that if $p_t = \mathcal{P}(a_t)$, all sectors have inflation equal to conflict inflation:

$$\pi_{nt} = \Pi_t^C, \quad n = 1, 2, \dots, N.$$

- Recall the ODE for nominal prices,

$$\pi_t = \Lambda (a_t - AP_t)$$

Subtracting conflict inflation on both sides gives

$$\pi_t - \mathbf{1}\Pi_t^C = \Lambda (a_t - Ap_t) - \mathbf{1} \frac{1}{d} \gamma' a_t$$

where $\mathbf{1}$ is a vector of ones, and where we use the fact that the rows of M sum to 1, which implies $AP_t = Ap_t$.

Proof of Result 2 - Continued

Given any a , define the relative price vector $\mathcal{P}(a)$ as the vector of relative prices p that satisfies

$$\Lambda(a - Ap) - \frac{1}{\bar{d}}\gamma' a = 0,$$

or, equivalently,

$$Ap = a - d\frac{1}{\bar{d}}\gamma' a$$

where $d = (d_1, d_2, \dots, d_N)'$. Let us prove that this vector exists and is unique. Recall from the proof of the Lemma that

$$\{y = AP \text{ for some } P \in R^N\} = \{y : \gamma' y = 0\}.$$

Then, given that

$$\gamma' \left(a - d\frac{1}{\bar{d}}\gamma' a \right) = \gamma' a - \frac{\gamma' d}{\bar{d}}\gamma' a = 0,$$

there exists a P such that

$$AP = a - d\frac{1}{\bar{d}}\gamma' a$$

Proof of Result 2 - Continued

- Letting $p = P - \mathbb{1} \sum \psi_n P_n$ and using again the fact that $AP = Ap$, we have a vector of relative prices that satisfies

$$Ap = a - d \frac{1}{\bar{d}} \gamma' a$$

- Uniqueness follows from A having rank $N - 1$.
- Given the definition of $\mathcal{P}(a)$, it follows immediately from

$$\pi_t - \mathbf{1} \Pi_t^C = \Lambda (a_t - Ap_t) - \mathbb{1} \frac{1}{\bar{d}} \gamma' a_t$$

that if $p = \mathcal{P}(a)$ then $\pi_{nt} = \Pi_t^C$ for all n .

Proof of Result 3

- From the system of differential equations for the relative prices, we have:

$$\dot{p}_t = \Lambda a_t - \mathbb{1} \frac{1}{d} \gamma' a_t - \Lambda A p_t$$

So

$$p_t = e^{-\Lambda A t} p_0 + (\Lambda - \mathbb{1} \frac{1}{d} \gamma') \int_0^t e^{-\Lambda A(t-s)} a_s ds$$

- The boundedness of $\{a_t\}$ implies that p_t is bounded.
- Define the relative price adjustment component of inflation as $\Pi_t^A \equiv \dot{p}_t$. Then inflation satisfies

$$\pi_t = \Pi_t^A + \Pi_t^C.$$

- $\dot{p}_t = \Pi_t^A$ implies

$$\frac{1}{T} \int_0^T \Pi_t^A dt = \frac{p_T - p_0}{T}.$$

- Since the numerator is bounded $\implies \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \Pi_t^A dt = 0$. □